

Answer Key & Teacher Notes

Session 1 – Estimate with Benchmarks • Session 2 – Estimate a Percent of a Number

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Estimate with Benchmarks

TALK PROMPTS

- Why does naming the whole first change the percent that you estimate?
- How does a benchmark let you judge an answer without doing exact arithmetic?

PARTNER TASK

Elena's Drive. A drive holds 200 GB; 150 GB are used, and 30 GB of that is music. One partner estimates the percent of the drive used (75%); the other estimates the percent of the used space that is music (30 of 150, 20%). Then explain why the wholes differ.

WATCH FOR

- Using the wrong whole (calling 30 GB of music 75% because 150 of 200 GB is used).
- Writing the part as the percent (47 out of 64 called "47%").
- Choosing a benchmark without comparing the part to the benchmark amount (19 out of 60 called 50%).

SESSION 2 • TEACHER NOTES

Estimate a Percent of a Number

TALK PROMPTS

- Why can rounding both the percent and the total still give a useful estimate?
- Does your estimate run high or low? Point to the rounding that tells you.

PARTNER TASK

Tip Estimate and League Roster. One partner rounds and names the friendly numbers; the other builds the estimate from 10% and says whether it runs high or low. Switch roles with Field Trip Coupon: about 19% of \$237 (about \$48).

WATCH FOR

- Misplacing the decimal when finding 10% (10% of 300 written as 3).
- Multiplying by 20 instead of 0.20 for 20% ($20 \times 500 = 10,000$).
- Writing an estimate with "=" and never checking whether it is high or low.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3c	1	2	1	1	2	2	1	2	3	1	1
S1 • Form B	6.RP.A.3c	1	2	1	1	2	2	1	2	3	1	1
S2 • Form A	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.RP.A.3c	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

estimate: A close answer that is not **exact**.

benchmark percent: An **easy** percent to use, like 10%, 25%, 50%, or 75%.

the whole: The full amount. It counts as **100%**.

- First name the **whole**. That amount is 100%.
- Find benchmark amounts: 50% is **half** of the whole, 25% is one fourth, and 10% is the whole $\div$ 10.
- Pick the benchmark amount that is **closest** to the part.

Example: The whole is **80 MB**. 50% of 80 MB is **40 MB**. 41 MB is just a little more than 40 MB. Answer: About **50%** of the download is done.

YOU TRY

1. a. 25% b. 75% c. 10%
2. GB: 16, 32, 48. 47 is close to 48, so about 75%.

EXIT TICKET

1. a. 50% b. 75%
2. 25% of 300 = 75, and 74 is close to 75, so about 25%.
3. **C** (50%)

PRACTICE FORM A

1. a. 50% b. 25% c. 50% d. 75% e. 10% f. 100%
2. a. Pages: 60, 120, 180 b. 118 is close to 120, so about 50%
3. a. \$40 b. 75% of \$400 = \$300, and \$310 is close, so about 75%
4. a. 16 cups b. 17 is close to 16, so about 50%
5. a. One fourth of 36 is 9; $27 = 3 \times 9$, so 75% b. 25%
6. No. 50% of 60 is 30, far from 19. 10% of 60 = 6, so 30% of 60 = 18. 19 is close to 18, so 19 out of 60 is about 30%.
7. 75% of 80 = 60, and 61 is close to 60, so about 75% are taken.
8. a. 10% of 480 = 48, so about 10% b. about 90%
9. 75% describes the used space (150 of 200 GB), not the music. Music is 30 of 200 GB: 10% of 200 = 20 GB, so 30 GB is about 15% of the drive.
10. **B** (one tenth)
11. **B** (25%)

Self-check word NEAREST

PRACTICE FORM B

1. a. 50% b. 25% c. 75% d. 10% e. 100% f. 50%
2. a. Pages: 80, 160, 240 b. 162 is close to 160, so about 50%
3. a. \$60 b. 75% of \$600 = \$450, and \$452 is close, so about 75%
4. a. 24 cups b. 23 is close to 24, so about 50%
5. a. One fourth of 40 is 10; $30 = 3 \times 10$, so 75% b. 25%
6. No. 50% of 40 is 20, far from 11. 25% of 40 = 10, and 11 is close to 10, so 11 out of 40 is about 25%.
7. 75% of 120 = 90, and 89 is close to 90, so about 75% are taken.
8. a. 10% of 360 = 36, so about 10% b. about 90%
9. 75% describes the used space (300 of 400 GB), not the photos. Photos are 40 of 400 GB: 10% of 400 = 40 GB, so photos are about 10% of the drive.
10. **B** (one half)
11. **B** (25%)

Self-check word PERCENT

GUIDED NOTES

estimate: An answer that is **close** to the exact amount.

compatible numbers: Numbers that are **easy** to compute with in your head.

ten percent: One tenth of the whole: divide by **10**.

- Round the percent and the whole to **friendly** numbers.
- Find **10%** first, then build: 20% is double, 5% is half of 10%.
- Write “about” because the answer is an **estimate**.

Example: 312 is about **300**. 10% of 300 is **30**. 20% is two 10% parts: $30 + 30 = 60$. Answer: About **60** bottles

YOU TRY

1. a. about 40 b. about 30 c. about 70
2. Round to \$40: $10\% = \$4$, $5\% = \$2$, so about \$6.

EXIT TICKET

1. a. about 10 (20% of 50) b. about 30 (75% of 40)
2. 40% of \$60: $10\% = \$6$, so $40\% = \text{about } \$24$.
3. **C** (80)

PRACTICE FORM A

1. a. about 20 (10% of 200) b. about 30 (20% of 150) c. about 20 (25% of 80) d. about 300 (50% of 600) e. about 27 (30% of 90) f. about 90 (75% of 120)
2. a. 50% of 180 = about 90 lb b. 10% of 180 = about 18 lb (10% of 200 = 20 lb is also reasonable)
3. Round to \$60. $10\% = \$6$, $5\% = \$3$, so the tip is about \$9.
4. $25\% \text{ of } 600 = 600 \div 4 = \text{about } 150$ players.
5. 20% of \$240: $10\% = \$24$, so $20\% = \text{about } \$48$.
6. No. 9% is about 10%, and 10% of 420 is 42. So 9% of 418 is about 42, not 200 (200 is about half of 418).
7. $75\% \text{ of } 160 = 3 \times 40 = \text{about } 120$ plants. High, because both numbers were rounded up.
8. a. $10\% \text{ of } \$80 = \text{about } \8 b. $4 \times \$8 = \text{about } \32 c. $\$80 - \$32 = \text{about } \$48$; a little high, because \$78.90 was rounded up to \$80
9. His rounding was fine, but 10% of 300 is 30, not 3. So $20\% \text{ of } 300 = 60$; 19% of 285 is about 60 items.
10. **B** (25)
11. **C** (200)

Self-check word **ROUNDED**

PRACTICE FORM B

1. a. about 30 (10% of 300) b. about 50 (20% of 250) c. about 40 (25% of 160) d. about 200 (50% of 400) e. about 18 (30% of 60) f. about 60 (75% of 80)
2. a. 50% of 240 = about 120 lb b. 10% of 240 = about 24 lb
3. Round to \$50. $10\% = \$5$, $5\% = \$2.50$, so the tip is about \$7.50.
4. $25\% \text{ of } 400 = 400 \div 4 = \text{about } 100$ kittens.
5. 20% of \$60: $10\% = \$6$, so $20\% = \text{about } \$12$.
6. No. 8% is about 10%, and 10% of 500 is 50. So 8% of 512 is about 50, not 400.
7. $75\% \text{ of } 120 = 3 \times 30 = \text{about } 90$ plants. Low, because both numbers were rounded down.
8. a. $10\% \text{ of } \$60 = \text{about } \6 b. $3 \times \$6 = \text{about } \18 c. $\$60 - \$18 = \text{about } \$42$; a little high, because \$59.80 was rounded up to \$60
9. Lee stopped at 10%. 25% is one fourth, so 25% of $400 = 400 \div 4 = 100$. 25% of 396 is about 100.
10. **B** (84)
11. **C** (90)

Self-check word **COMPUTE**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • ESTIMATE WITH BENCHMARKS

1. Pages: 40, 80, 120. 42 is close to 40, so about 25%.
2. a. 10% b. 75% c. 25% d. 50% e. 100% f. 25% g. 10% h. 75%
3. a. \$50 b. 20% of 500 = \$100, and \$96 is close to \$100, so about 20%.
4. 75% of 48 = 36, and 35 is close to 36, so about 75%.
5. 26 out of 50: half of 50 is 25, so it is just over 50% (exactly 52%). 26 out of 60: half of 60 is 30, so it is below 50% (about 43%). 26 out of 50 is closer to 50%.

SHEET 2 • ESTIMATE A PERCENT OF A NUMBER

1. a. about 6 (10% of 60) b. about 30 (75% of 40) c. about 50 (25% of 200)
2. a. \$20 b. about \$60 ($3 \times \$20$)
3. 50% of 400 = 200, so about 200 students.
4. 25% of 800 = $800 \div 4 = 200$, so about 200 apples.
5. 20% is not 20; she multiplied by 20 instead of 0.20. 10% of 500 = 50, so 20% of 500 = 100. 19% of 512 is about 100.
6. **B** (20)
7. **C** (\$48)